Maschinelles Sehen

Course Notes

WiSe 25/26

Image Processing Pipeline

3D Scene → Image Formation Process → Image Representation

3D -> 2D

- Sensorqualität
- Blendeneinstellungen
- Linsenverzerrungen

⇝ Kalibrierung

Resolution → Aliasing

- png/jpeg
- RGB
  - ↳ Farbräume
- LAB
- Freq. Raum

→ Image Features → Applications

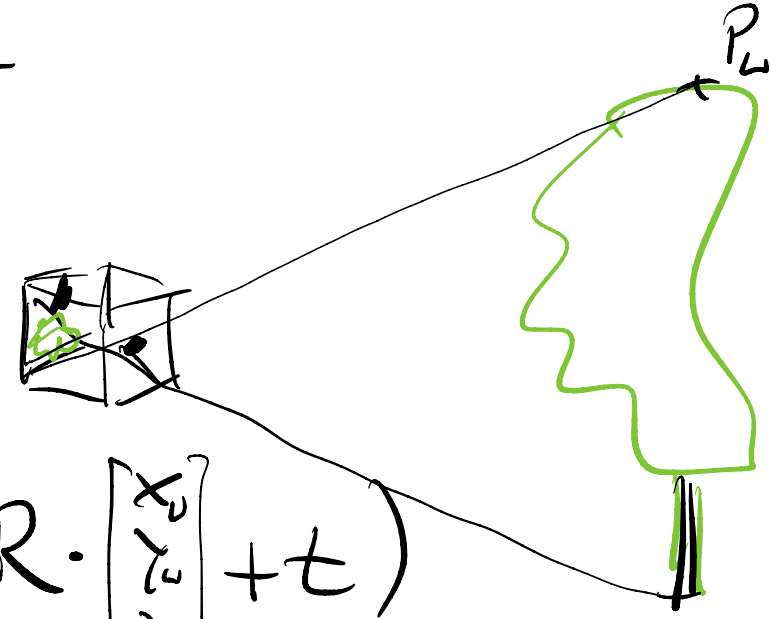
- edges, corners
- gradienten
- Histogramm

$$\vec{v} = \begin{bmatrix} 0.2 \\ 0.7 \\ \vdots \\ \vdots \end{bmatrix} \quad \mathbb{R}^N > 100$$

- Bildsuche
- 3D-Rekonstruktion
- Segmentierung

⋮

Image Formation



$$\begin{bmatrix} x_p \\ y_p \\ 1 \end{bmatrix} = \boxed{K}\left(R \cdot \begin{bmatrix} x_w \\ y_w \\ z_w \end{bmatrix} + t\right)$$

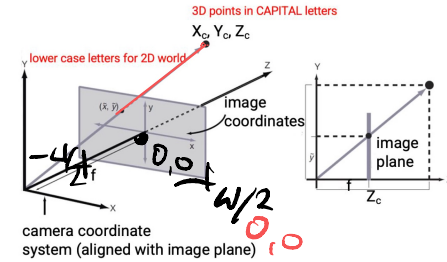
homogene Koordinate

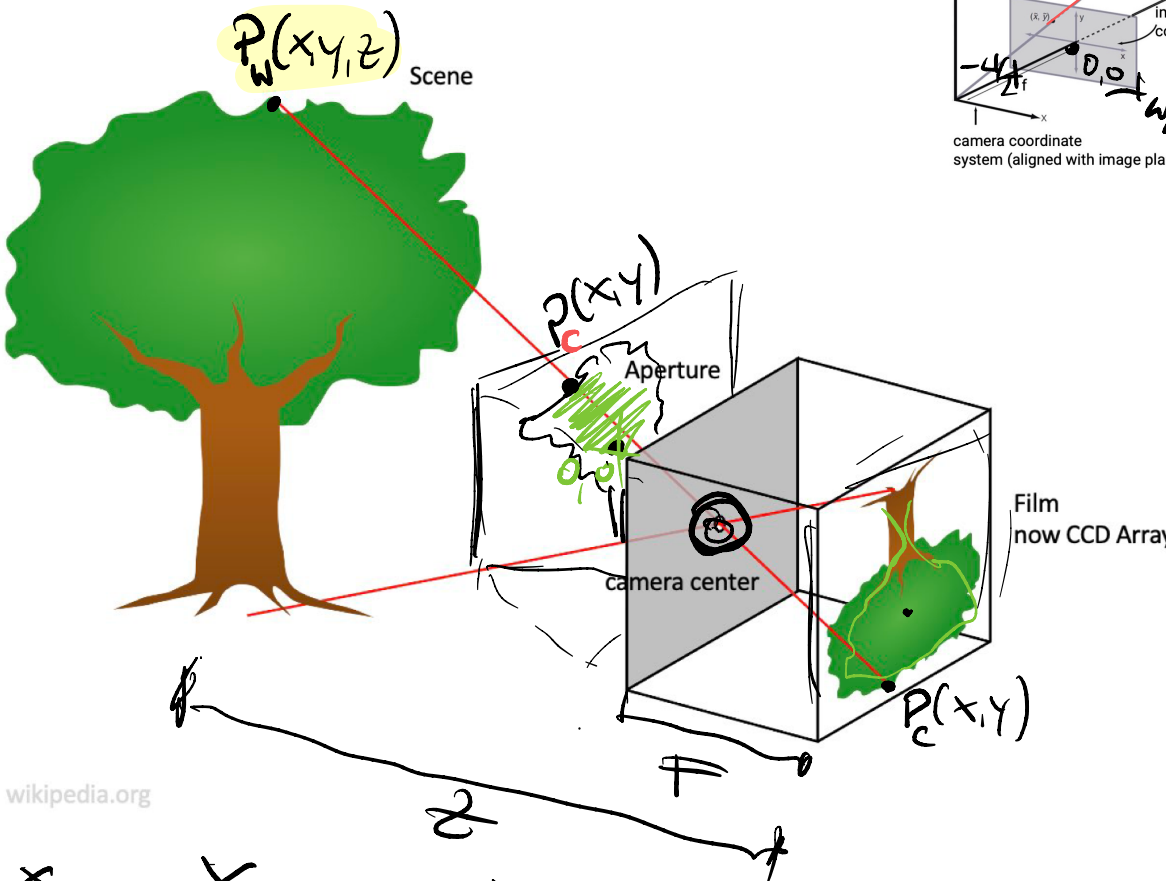
K – Camera Matrix

→ intrinsic camera parameters

→ extrinsic camera parameters

Image Formation





wikipedia.org

$$\frac{x_c}{f} = \frac{x_w}{z_w} \qquad \frac{y_c}{f} = \frac{y_w}{z_w}$$

$$x_c = F \cdot \frac{x_w}{z_w}$$

physical location
CMOS
CCD
mm

mm

m
mm

$$\alpha_x = \frac{F}{dx}$$

$$\alpha_y = \frac{F}{dy}$$

$$x_p = \frac{x_c}{dx} + x_0$$

dx → wie groß ist ein Pixel?

Camera Matrix K

$$K = \begin{bmatrix} \alpha_x & 0 & x_0 \\ 0 & \alpha_y & y_0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x_w \\ y_w \\ z_w \end{bmatrix} = \begin{bmatrix} x_p \\ y_p \\ 1 \cdot z_w \end{bmatrix} \Big/ z_w$$